SFWRENG/MECHTRON 4AA4 - Real-Time System and Control Applications

Course Syllabus - Fall Term 2023

Instructor:

Dr. Zhao Zhao (first half of the course, before Reading Week)

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Office hours: TBA

Teaching Assistants:

Zikai Dou, email: douz7@mcmaster.ca (Lab 3 and 8)

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Akshobhya Katte Madhusudana, email: madhusua@mcmaster.ca (Lab 6 and 7)

Lectures and Labs:

- Lecture:
  - Mo 11:30AM - 12:20PM MDCL 1105
  - We 11:30AM - 12:20PM PGCLL M21
  - Fr 1:30PM - 2:20PM PGCLL M21
- Lab (starting from Sep 11) in ITB 235:
  - Mo 8:30AM - 11:20 AM
  - Tu 2:30 PM - 5:20 PM
  - Th 2:30PM - 5:20 PM
  - Fr 8:30 AM - 11:20 AM
  - Fr 2:30 PM - 5:20 PM

Reference book

Buttazzo, G. C. (2011). Hard real-time computing systems: predictable scheduling algorithms and applications (Third Edition). Springer Science & Business Media.

Available Online

Evaluation

- 25% Midterm Exam [To be confirmed, preliminary: Oct 6 in class]
- 35% Final Exam [To be confirmed]
- 40% Labs

Note: bonus marks will be given for in-class quizzes and participation.

Intended Learning Outcomes

This course is designed for senior-level undergraduate students who have the necessary background in operating systems and physical system control. This course will cover two topics: (1) the design and analysis of real-time systems and (2) the design and analysis of control systems.

The former is about the design of real-time operating systems. We will learn basic real-time operating systems concepts and services, including interrupt processing, process and thread models, and especially scheduling algorithms. The outcomes of this course are:

- Understand the differences between general purpose and real-time operating systems.
- Understand task and thread scheduling in real-time operating systems.
- Understand memory management in real-time systems.

The latter emphasizes how a continuous control system can be approximated to a digital control system and how to implement a controller using software running on digital computers. Besides the lectures, 10 lab sessions are designed to help students gather hands-on experience in the design of real-time systems and control applications.

Tentative Schedule

The intended schedule and corresponding textbook chapters are listed in the table below:

Session	Contents
Week 1, 06/08 Sep	**Introduction, Basics of RTOS**
	Buttazzo Chapter 1 and 2
Week 2, 11/13/15 Sep	**Aperiodic task scheduling**
	Buttazzo Chapter 3
Week 3, 18/21/22 Sep	**Periodic task scheduling**
	Buttazzo Chapter 4
Week 4, 25/27/29 Sep	**Resource access protocols**
	Buttazzo Chapter 7
Week 5, 02/04/06 Oct	**Limited preemptive scheduling**
	Buttazzo Chapter 8
	Midterm - Oct 6 in class

Week 6, 9/11/13 Oct	**Midterm recess**
Week 7, 16/18/20 Oct	**TBD**
Week 8, 23/25/27 Oct	**TBD**
Week 9, 30 Oct 1/3 Nov	**TBD**
Week 10, 06/08/10 Nov	**TBD**
Week 11, 13/15/17 Nov	**TBD**
Week 12, 20/22/24 Nov	**TBD**
Week 13, 27/29 Nov 1 Dec	**TBD**
Week 14, 04/06 Dec	**TBD**

Formal conditions

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty. Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/ The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGIARISM DETECTION

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty. Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE ELEMENT

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities(the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online. It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact Student Accessibility Services(SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.cato make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilitiespolicy.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

In the event of an absence for medical or other reasons, students should review and follow the Policy on Requests for Relief for Missed Academic Term Work.

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISOpolicy. Students should submit their request to their Faculty Office normally within 10 working days of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

COPYRIGHT AND RECORDING

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EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.